

Jinay Patel

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EDUCATION

University of Waterloo

Bachelor of Computer Engineering

Expected Graduation: Jun 2031

SKILLS

Languages: C/C++ (5 years), JavaScript/TypeScript (6 years), Python (5 years), Java (2 years)

Technologies: PyTorch, Langchain, Mastra, CMake, React, Svelte, Next JS, Tailwind CSS, Git, Docker, Unity

EXPERIENCE

Lead Software Engineer

October 2023 — June 2026

Seaquam Robotics

Delta, BC

- Built a data backend with Deno and TypeScript to aggregate data from 3 sources, cutting manual data entry by +50%
- Created a custom React and Tauri CMS interface, reducing data entry time from weeks to 60 minutes
- Built a Deno backend caching layer for slow third-party API endpoints, increasing server response speed by +150%

Embedded Controls Software Engineer

October 2021 — June 2026

Seaquam Robotics

Delta, BC

- Implemented C++ motion algorithms, securing a 100% win rate during autonomous periods across 15+ matches
- Designed an Interpolated Look Up Table using a Monotonic Cubic Spline for tuning PID constants in real-time which resulted in a 200% increase in accuracy across all movements with error tolerances being <1 inch

Software Engineering Tutor

September 2025 — March 2026

Code Ninjas

Surrey, BC

- Helped 5 senior Black Belt students debug and implement features for their games being built with C# and Unity
- Designed ad campaigns with visual assets for Code Ninjas, improving our local branch search visibility

Freelance Systems Administrator / IT Consultant

Convergence Media

Remote

- Created a custom Linux Mint ISO using Cubic with 3 bash scripts to automate the installation and setup of programs
- Deployed a self-hosted Nextcloud server (Microsoft 365 alternative) using Docker, which allowed file sharing and private video calls
- Configured a Tailscale tunnel to expose a public link and allow anyone to access the Nextcloud server for video calls or file sharing

PROJECTS

Motion Control Framework | Embedded Controls Software Engineer

[GitHub](#)

- Used C++ to make a robot move autonomously using PIDs, feedforward controllers, and 2D motion profiles
- Designed a custom 2D Motion Profile generator which used Cubic Bezier curves for calculating the robot's trajectory at each point on the path, and then compensating for centripetal forces to allow the robot to stay on the path
- Developed visualization software using Svelte and Websockets to view position and PID data in real time

Verbosity: Custom Programming Language | Personal Project

[GitHub](#), [Blog Post](#)

- Developed a custom lexer, parser, and code generator for a programming language using C++ and CMake
- Applied C++26 features such as abstract classes, virtual methods, smart pointers, and variants

Termicord - Real-Time CLI Chat Application | Personal Project

[GitHub](#)

- Developed a Discord-like chat application using C++, raw TCP sockets, NCurses (for the TUI), and VCPkg
- Implemented a custom HTTP server with routing that allows for requests to be sent to a REST API

AWARDS

Excellence Award @ Vex Robotics World Championship, RECF & Vex Robotics

April 2026

Awarded to the top team out of 20,000+ teams worldwide for excellent performance

Excellence Award @ Provincial Championship, Vex Robotics

February 2026

Awarded to the top team out of the 100+ teams in British Columbia for excellent performance

VSHacks Hackathon 2nd Place, VSHacks

July 2025

Awarded for developing [Posturai](#), a real-time posture correction software

Euclid Math Contest Distinction, Waterloo CEMC

May 2026

Top 25% in Canada for the Euclid math contest